

Prerequisite: Database Management Systems /Computer Networks

Course Objectives

The course on Cloud Computing And Big Data Analytics aims to provide the students with the following:

1. Understand various basic concepts related to cloud computing technologies
2. Understand the architecture and concept of different cloud models: IaaS, PaaS, SaaS.
3. Be familiar with application development and deployment using cloud platforms.
4. Explore the fundamental concepts of big data analytics
5. Understand the Hadoop architecture and Ecosystem components along with Mapreduce.

Course Outcomes

On successful completion of the course, the student will be able to:

1. Identify and select the cloud services with different service models
2. Display their knowledge on Virtualization and cloud data center architecture.
3. Outline different methods to access cloud and various techniques to collaborate within cloud.
4. Correlate their knowledge of problems associated with big data in various domains.
5. Interpret the features of Hadoop architecture and run Map reduce applications.
6. Identify the applications of Cloud services, Big data analytics and Hadoop architectures
7. Associate the resource provisioning in the cloud services using VMware workstation and CSP
8. Relate suitable Cloud deployment services and Big Data analytics to find solution to real time problems.

UNIT I CLOUD AND SERVICES

Introduction – Evolution of cloud computing – Hardware evolution – Internet software evolution – Server virtualization – Web services overview - IaaS - PaaS – SaaS - XaaS.

UNIT II VIRTUALIZATION FUNDAMENTALS

Virtualization architecture- Hypervisors- Security problems in Virtualization- Cloud Datacenter architecture- Software defined datacenter- Service Oriented Architecture – Cloud Federation – Presence – Identity Privacy.

UNIT III ACCESS TO CLOUD

Hardware and infrastructure – Clients – Security – Network – Services - Accessing the cloud – Platforms – Web applications – Web APIs- Web browsers. Cloud storage overview – Storage Area Network Architecture- Collaboration within the cloud – Standards –Driving forces – Software and services – Developing applications.

UNIT IV FUNDAMENTALS OF BIG DATA

Introduction - Distributed file system - Structured and unstructured data- Big data and its importance - 5Vs - Drivers for Big data - Big data analytics - Data appliances and Integration tools - Big Data Security - Issues in Big data - Big data applications.

UNIT V HADOOP AND MAPREDUCE ARCHITECTURE

Big data – Apache Hadoop & Hadoop EcoSystem – Analyzing data with Hadoop streaming – HDFS concept – Interface to HDFS -Moving Data in and out of Hadoop –Introduction to MapReduce – MapReduce Algorithm and Architecture – Understanding inputs and outputs of MapReduce - Anatomy of MapReduce Job run – Failures in classical MapReduce and YARN – Job scheduling - Data Serialization.

TEXT BOOKS

1. Rittinghouse John, and Ransome James, “Cloud Computing: Implementation, Management, and Security”, CRC Press, 2010.
2. Toby Velte, Anthony Velte and Robert Elsenpeter, “Cloud Computing - A Practical Approach”, Tata McGraw Hill, 2010.
3. Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, “Professional Hadoop Solutions”, Wiley, 2015.
4. Chris Eaton, Dirk deroos et al. , “Understanding Big data ”, McGraw Hill, 2014.

REFERENCES

1. Michael Miller, “Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online”, QUE publishing, 2009.
2. George Reese, “Cloud Application Architectures: Building Applications and Infrastructure in the cloud”, O Reilly Publications, 2009.
3. Miller F.P., Vandome A.F and McBrewster John., “Cloud Computing”, Alphascript Publishing, 2009.
4. Vignesh Prajapati, “Big Data Analytics with R and Hadoop”, Packet, 2013.